

# QUALITATIVE AND MIXED-METHODS RESEARCH IN ECONOMICS: SURPRISING GROWTH, PROMISING FUTURE

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**Abstract.** Qualitative research in economics has traditionally been unimportant compared to quantitative work. Yet there has been a small explosion in use of quantitative approaches in the past 10–15 years, including ‘mixed-methods’ projects which use qualitative and quantitative methods in combination. This paper surveys the growing use of qualitative methods in economics and closely related fields, aiming to provide economists with a useful roadmap through major sets of qualitative methods and how and why they are used. We review the growing body of economic research using qualitative approaches, emphasizing the gains from using qualitative- or mixed-methods over traditional ‘closed-ended’ approaches. It is argued that, although qualitative methods are often portrayed as less reliable, less accurate, less powerful and/or less credible than quantitative methods, in fact, the two sets of methods have their own strengths, and how much can be learned from one type of method or the other depends on specific issues that arise in studying the topic of interest. The central message of the paper is that well-done qualitative work can provide scientifically valuable and intellectually helpful ways of adding to the stock of economic knowledge, especially when applied to research questions for which they are well suited.

**Keywords.** Mixed-methods; Qualitative methods; Survey methodology; Survey research

## 1. Introduction

Qualitative research in economics has traditionally been relatively unimportant compared to quantitative work. Doctoral programs provide 2–4 semesters in statistics and econometrics, but rarely even mention qualitative methods. Research using qualitative approaches – such as in-depth interviews, focus groups and case studies – is occasionally published in field journals and used in interdisciplinary work; on rare occasions, studies based on qualitative evidence achieve major influence in the profession (Bewley, 1995, 1999; Blinder *et al.*, 1998). But for the most part, qualitative methods are not thought of as being part of the economist’s toolkit (Berik, 1997; Bewley, 2002; Dow, 2007).

Yet the past 10–15 years have seen a small explosion in use of quantitative approaches in specific fields of economic research, including ‘mixed-methods’ research projects which use qualitative and quantitative methods in combination. Research areas in which such methods are increasingly used include: studies designing or gauging the effects of social programs, especially among lower income groups; studies of willingness to pay for environmental interventions; studies related to poverty and capabilities sponsored by the World Bank; case-study research into innovation, R&D, and technological diffusion; and feminist-economics research into the ‘lived experiences’ of women’s economic lives. Much of this research is quite unlike economists’ impression of what qualitative research amounts to, i.e. it is rigorous and carefully conceptualized, with considerable effort made to tackle issues of potential biases in what information is collected and what inferences are drawn from it.

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This paper surveys this growing use of qualitative methods in economics and closely related fields. A key objective of the paper is to provide economists with a clear and helpful roadmap through major sets of qualitative methods, with the intention of expanding the profession's understanding of scientifically valuable approaches to building economic knowledge. Important here is the fact that, although qualitative methods are often portrayed as less reliable, less accurate, less powerful and/or less credible than quantitative methods, in fact, the two sets of methods have their own strengths, and how much can be learned from one type of method or the other depends on specific issues that arise in studying the topic of interest – including the nature of the subject being studied, what else is already known about that subject, priorities in terms of gaps in knowledge that need to be filled, and aspects of the design and implementation of the study that bolster confidence in the inferences that can be drawn from it.

The next section of the paper provides basic definitions and characterizations of different types of qualitative and mixed-methods research of particular relevance for economic research. An important argument made here is that, although economists often think of qualitative research as involving words and quantitative research as involving numbers, a more valuable way of thinking of the distinction is in terms of open- vs. closed-end approaches to gathering data. Thus, in quantitative studies, researchers gather or use data with the expectation that they know in advance a fixed set of dimensions along which the data should be characterized (e.g. income level, daily share prices, given instruments of monetary policy); in qualitative studies, researchers 'proceed to the field' with clear and detailed guidelines as to what issues they want to investigate and how, but expecting their *interaction* with research subjects and/or their broad review of relevant data to provide the basis for constructing a sound characterization of the phenomena of interest. The third section discusses basic issues in qualitative research, including types of qualitative methods, issues in sample size and composition, and methods for analyzing qualitative data. The fourth section reviews the surprisingly large body of economic studies using qualitative approaches, emphasizing the gains from using qualitative approaches or methods that mix qualitative and quantitative approaches over closed-ended types of research methodologies. The fifth section discusses some of economists' general concerns about the 'objectivity' of qualitative research, rooted in the idea that, because 'talk is cheap', one must study what people *do* and not what they say. It is argued that many concerns raised here are methodological problems that can be tackled through careful research design, training of interviewers, balanced interpretation of evidence, etc. – rather than constituting grounds for discounting the value of open-ended research approaches *a priori*. The final section argues that, because qualitative methods provide good opportunities to bring the perspectives, experiences and understandings of research subjects into processes of producing economic knowledge, they are valuable not only for improving the scientific validity of economic knowledge, but also the ethical properties of its production and social value.

## 2. Distinctions between Qualitative and Quantitative Research

Economists typically think of the distinction between quantitative and qualitative research as being that the former analyzes numerical data using statistical or econometric methods, while the latter uses data expressed in words and analyzed some other way. While fine for telegraphing the distinction, this differentiation is too simple as a basis for a review of the growing use of qualitative methods in economic research.<sup>1</sup> For one, the world of information types is not starkly and completely divided into 'numerical' and 'verbal' categories, with all batches of information uniquely belonging in one category or the other. It can be straightforward to transfer data of one kind into the other: for example, survey respondents are often asked to express qualitative views along a quantitative scale (as in surveys of business or consumer confidence), and researchers often convert narrative information into quantitative forms by using numerical codes (as with studies of central bank decision-making and

communications). For another, there is not a one-to-one correspondence between the nature of the data and the type of method used to analyze it, as when quantitative evidence is analyzed narratively (Friedman and Schwartz, 1963) or qualitative material is transformed so it can be analyzed using statistical or econometric methods (Starr, 2012). Finally, with economists more familiar with and predisposed towards quantitative approaches, the profession has some tendencies to associate 'good' research qualities (representativeness, rigor, objectivity, explanation) with quantitative data and methods, and corresponding 'bad' ones (un-representativeness, informality, subjectivity, description) with the qualitative. Yet both 'good' and 'bad' research qualities occur in both types of research; for example, econometric analysis of data from a casually drawn 'convenience' sample and subject to a lot of measurement error may or may not permit better inferences than systematic analysis of carefully recorded and coded 'verbal' data drawn from a well-constructed purposive sample. All that may differ here, as Helper (2000) points out, is that economists know how to deal with problems that complicate inference in quantitative work (measurement error, missing data, self-selection, etc.), but have no basis for thinking about such problems in qualitative work.

If the key distinction between qualitative and quantitative research is not words versus numbers *per se*, what is? The primary difference between the two, which gives rise to several others, concerns the open-ended character of data collection in qualitative research. In standard quantitative research, a pre-determined set of information items is collected from research subjects (e.g. respondents to surveys) or data-reporting units (e.g. companies filing quarterly financial reports, meteorological stations reporting weather data, etc.), where the only information collected is what has been pre-specified in the research instrument.<sup>2</sup> Research subjects cannot question the questions they are asked, add nuances or caveats, or explain the reasoning behind their response. Instead it is assumed *a priori* that the researcher knows the specific informational items that played a central role in the subjects' behaviours, perceptions and/or decisions, and can compellingly hypothesize how these items interrelate. In contrast, in qualitative studies, the approach to information gathering assumes that relatively flexible discussions with research subjects are needed for gaining a full and complete set of insights into the phenomenon of interest. Thus, for example, whereas a closed-end survey of home-buying behaviour may ask 'Please tell me which of the following best describes your main reason for buying this house', where the respondent is offered a fix set of answers to choose from [e.g. '(a) good investment, (b) ready to start a family, (c) location is close to work/school/family, etc.'], a qualitative study may instead ask the respondent to 'Tell me how you came to buy this house'. Respondents then explain their experiences in their own words, possibly with the interviewer asking clarifying questions and/or 'probing' on questions of research interest. But the goal is to recover a full picture of the factors and processes (e.g. cognitive, social, informational) at work in the respondent's thinking, as well as the opportunities and constraints present in the environment that shaped his perceptions, beliefs and behaviours (e.g. borrowing constraints).

Obviously an open-ended approach to data collection yields more information than the closed-end one, although its unstructured character means that what comes in varies from subject to subject, making analysis of the data less straightforward. While the availability of software programs like NVivo, MAXQDA and Atlas.ti has significantly improved the opportunities for analyzing narrative information in systematic ways, the information collected via qualitative methods is inevitably richer (more detailed, more complex, having less *a priori* structure) than what is collected in quantitative work. Thus, especially in research undertaken by economists, these methods tend to be used only when the extra richness of the information clearly warrants the greater complexity of its analysis. There are well-defined circumstances under which this may be the case; examples of economic studies motivated by these rationales are discussed in Section 4. But to lay these out in general, they include circumstances: (a) when very little is known about the topic, so that broad exploratory research is needed to identify its basic characteristics; (b) when there has already been a lot of quantitative research on the subject, but key questions remain unresolved; (c) when back-and-forth with an interviewer is thought to be needed to help elicit full and accurate information; (d) when the topic under investigation has some

inherent complexities the researcher wants to be able to capture; and/or (e) when respondents' own views of their own situations are of inherent interest.

### 3. Basics of Qualitative Research

#### 3.1 *Types of Qualitative Methods*

There are five main ways of collecting qualitative data that are especially relevant to economic research. It should be stressed at the outset that they are increasingly used in combination, either with each other or with quantitative approaches, as a matter of 'triangulation' – that is, cross-checking the accuracy, validity, relevance and completeness of the information coming from the research against information from other sources.<sup>3</sup> In this regard, we also consider a sixth booming category of 'mixed-methods' research, which specifically uses both qualitative and quantitative approaches together as a matter of exploiting the strengths of each – namely, depth and complexity on the qualitative side, vs. representativeness and statistical power on the quantitative.

1. *In-depth interviews.* In-depth interviews refer to extended discussions with research subjects. These may be 'structured', 'semi-structured', or 'unstructured', referring to the extent to which the conversation follows a pre-determined sequence of questions. Projects with large numbers of interviews tend to be semi-structured or structured, to ensure sufficient comparability across interviews in the information collected, while smaller scale projects or projects that give priority to recovering relatively unfiltered representations of respondents' own views tend to be lower on structure. Ideally, the interviews are taped and later transcribed, as this preserves the full information content of all interviews and facilitates automated cataloguing and analysis of the data. Some researchers just take notes and elaborate on them as soon as possible afterwards, which may be sufficient if only one person is interviewing, the interview is not too long, and the number conducted is not too large; sometimes it may be necessary when respondents do not want to be taped. In either case, preferred practice is to keep detailed, uniform records of interviews that can be consulted and analyzed systematically *ex post*. Bewley (1999) is a major economic study based primarily on in-depth interviews (see below).<sup>4</sup>
2. *Focus groups.* Focus groups are semi-structured group-discussion sessions, where a facilitator raises questions for participants to discuss, and predictable conversational dynamics (plus specific methods used by the facilitator) help bring out majority and minority perceptions, opinions, views and experiences within the group. Because discussions within specific groups may be influenced by unusual individuals or clusters thereof, it is preferable to run multiple groups to average out group-specific idiosyncrasies. Additionally, because discussions tend to be most open and productive when groups are relatively homogeneous, acquiring insights on different population segments is best done by including them in separate focus groups and comparing results across them. Focus groups are not much used as primary vehicles for data collection in economic research, but they have been used to round out insights from other types of methods (Kennickell, Starr-McCluer and Sundén, 1996; van Staveren, 1997; Chilton and Hutchinson, 1999).
3. *Case studies and site visits.* The case-study method involves using a relatively small number of cases (countries, communities, companies or individuals, depending on the research purpose) to conduct an in-depth analysis of a given question of interest. Detailed information is collected for each case, often using multiple sources; for example, case studies of businesses commonly interview executives and managers, compile and analyze the firms' financial records, collect Lexis-Nexis information on media coverage of news on the company or sector, etc. Then commonalities and differences in experiences across cases are used to establish key empirical patterns, develop new explanations for observed phenomena, and/or gauge the extent to which prevailing

theoretical understandings of the phenomena of interest are consistent with evidence provided by the cases. In economics, case studies have been a long-time staple of research in the field of industrial organization, especially because dynamic processes like research-and-development, adoption of new technologies, entrepreneurship, patterns of collaboration and competition, etc. are often hard to 'see' from standard firm-level data alone (e.g. on R&D or capital expenditures, employment, patents or sales). Within the category of 'case studies', economists have a special tradition of 'pin factory' visits – traced back to Adam Smith's observations on the 18 stages of making a pin, via which he laid out seminal ideas about the division of labor, productivity and income growth. Thus, economists have to varying degrees recognized the value of visiting the 'shop floor' as a means of gaining unexpected, new, empirically grounded insights into key economic processes.<sup>5</sup> Site visits are not usually a stand-alone research method but rather figure into broader research projects using case-study, interview and/or quantitative methods; see Borenstein, Farrell and Jaffe (1998) and Helper (2000) for discussion. Perhaps the most influential use of site visits was the year Ronald Coase spent traveling to US factories and businesses, talking to decision-makers and observing patterns of inter- and intra-business transactions; as discussed in Coase (1988), this material contributed centrally to the development of his understanding of horizontal and vertical integration, presented in his seminal 'Nature of the Firm' (1937).<sup>6</sup>

4. *Fieldwork or ethnography.* At the other end of the spectrum from pin-factory visits, ethnographic research entails extended observation of a given community or group, aiming to characterize the norms, rules, conventions, habits and beliefs that govern patterns of behaviour and interaction of its members. Full-fledged fieldwork requires 1–3 years of commitment to developing broad networks of relationships in the community and acquiring access to opportunities to participate in its core activities. While economists rarely conduct such full-fledged field work (see Berik, 1997), there is a growing body of anthropological, sociological and interdisciplinary research based on fieldwork that examines topics of economic interest, such as the culture of Wall Street investment banks (Ho, 2009), career ladders in urban gangs (Venkatesh, 2008), ethnographies of working- and middle-class life (Roberson, 1998), and understandings of consumption in low-income communities (Chin, 2001). Occasional collaborations between economists and other social scientists (e.g. Levitt and Venkatesh, 2000, 2001) illustrate the potential value of fieldwork as a means of investigating topics requiring sustained in-depth collection of information.
5. *Life histories.* Life histories refer to information collected via in-depth interviews with subjects about important events and periods over their lives. As such, they provide a way to gain direct insight into low-frequency, longitudinal processes that shape life outcomes; they are also valuable for capturing people's perceptions of periods of important social and economic change and how their lives were affected by them. As with ethnography, life histories are not much used by economists, although there are some interesting exceptions (Olson and Emami, 2002), and there is life-history work in related disciplines on economic topics. For example, in Buckland, Fikkert and Eagan (2010)'s study of very low-income Canadians, the life history approach helped to underline that, although respondents had intermittent periods of improving livelihoods and capabilities during their lives, these were hard to sustain due to the multiple disadvantages most faced (e.g. mental illness, substance abuse).
6. *Mixed-methods research.* Mixed-methods research covers a diverse set of practices for combining qualitative and quantitative methods, in the interests of exploiting the strengths of both types of research and offsetting each others' weaknesses.<sup>7</sup> Three main ways in which the two types of research are combined are: (a) conducting a first exploratory, qualitative phase, which is used to design a second quantitative phase intended to generalize results to the population; (b) first administering a large-scale survey, then following up with in-depth interviews or focus groups to round out and enrich the findings; and/or (c) fielding the two types of projects concurrently and analyzing and interpreting the data together, where insights into unexpected results from

one method can be gleaned from results of the other. Although usually referring to survey-type research, the quantitative dimension of mixed-methods research could also be analysis of administrative records or formal experiments.<sup>8</sup> Mixed-methods research has exploded in certain fields recently, especially evaluation of social programs and health-services research (see, e.g., London, Schwartz and Scott's, 2006 overview) and in research on poverty in developing countries (e.g. Bamberger, 2000; Kanbur and Shaffer, 2007); research in the latter area is called 'Q-squared', referring to gains that come from having both qualitative and quantitative evidence that ought to be 'squared'. Mixed-methods work is often conducted by interdisciplinary teams, in which other researchers skilled in qualitative methods offset economists' lack of background, as discussed further below.

### 3.2 *Sample Sizes and Compositions in Qualitative Research*

Because collection and analysis of qualitative data are resource-intensive, sample sizes are typically much smaller than in standard closed-end surveys; they also tend to be constructed 'purposively' rather than drawn randomly from a sample frame. Purposive sampling covers a wide variety of practices referring to the construction of the sample in some way that facilitates satisfaction of the research objectives. For example, the sample may be constructed to ensure that important dimensions of variation in the population are also present in the sample. Alternatively, it may contain sub-samples which permit hypotheses to be examined, as when one part of the sample has a trait or experience of interest and the other functions as a control group (see National Science Foundation, 2004).

It is a problem that, with economists schooled in the idea that 'more n is better', sample sizes considered to be sufficient in other fields (e.g. interviews with 25–30 people) are usually viewed as too small to be informative by economists. Indeed, even when the sample size gets very large by other fields' standards (e.g. Bewley's single-handed interviewing of 300+ businesspeople), the reaction of economists may still be that the sample is not large or diverse enough to permit inferences to be drawn for the population. While ability to draw inferences is always a concern in qualitative research, it is also true that sample size *per se* is not the central issue: because one chooses to do qualitative research for its different strengths – i.e. the opportunity for in-depth investigation – the question is whether the information and insights it generates succeed in advancing understanding of the topic by more than would have been possible from larger-n, closed-end research. It is nonetheless important to be clear about the basis for selecting who to interview or what cases to study, and to establish why they should be seen as relatively typical members of the groups of interest.

### 3.3 *Methods of Analyzing Qualitative Data*

When economists analyze qualitative data, they tend to focus on identifying clear patterns in the data using a 'reasonable person standard' – that is, aiming to conduct the same sort of careful, systematic analysis of the data that any reasonable member of the profession concerned with scientific validity would if he/she were in the researcher's shoes. If anything, there may be some tendency to err on the conservative side to avoid any semblance of mixing views of the researcher in with interpretation of the data. This narrowly focused approach contrasts with approaches taken in other fields, which tend to be intentionally inductive, acknowledge but do not try to erase the role of the researcher from the interpretation of the results, and may place priority on bringing out the views and experiences of the research subjects.<sup>9</sup> Some of these approaches can look problematic to economists, given the discipline's strong beliefs in deductive reasoning and objective measurement, and distrust of people's own explanations and descriptions of their thinking and behaviours (McCloskey, 1998; Helper, 2000; Bertrand and Mullainathan, 2001) – although, as will be argued below, one can see these beliefs as unreasonably blocking off sensible Bayesian-style approaches to making sense of data.



At the same time, there is active discussion in other fields about how to increase the rigor of analysis of qualitative data and improve the extent to which inferences about causalities can be drawn from qualitative work (e.g. Burgelman, 2009), including debates about how to build possibilities of hypothesis testing into qualitative projects (e.g. National Science Foundation, 2004). In the first area, as mentioned, the availability of software packages for analyzing narrative information (e.g. NVivo, MAXQDA and Atlas.ti) now enables researchers to rely less on impressions and recollections of discussions with research subjects, and more on transcribed records that can be compiled and analyzed electronically.<sup>10</sup> In the second, efforts to explore causalities include asking research subjects to think through ‘hypotheticals’ or participate in experiments, and/or structuring samples in ways that enable hypotheses to be examined. For example, if a given type of context A is hypothesized to give rise to the development of a given phenomenon B, then just as in quantitative work, qualitative information on differences between the samples in which context A is and is not found (in some plausibly exogenous way) would provide a way to ‘disconfirm’ the posited relationship. But even among qualitative researchers who share economists’ concern with isolating causal channels, the discussion tends to be far more accepting of possibilities of back-and-forth between theory and evidence than economists are accustomed to, as when implications of a theory are checked against the evidence, then the theory may be revised to account for non-conforming observations (NSF, 2004). We return to discussion of this issue in Section 5 below.

#### 4. Uses of Qualitative Research in Economics

This section reviews important bodies of research within economics or with a central economic focus in which qualitative or mixed-methods have been used to good effect. Table 1 provides summary information on notable studies in these areas, recording the question addressed, the methods used, the sample size and the specific gains from using an open-ended method over what would have been possible from a standard closed-end approach.

##### 4.1 *Wages, Prices and Macro Theory*

Bewley’s (1999) study of wages and Blinder *et al.*’s (1998) study of prices are two major pieces of economic research using qualitative methods. Bewley interviewed 300+ employers in Connecticut and nearby states after the 1990–91 recession, asking why they did not cut wages even when slack labor-market conditions would seem to imply that they could. Blinder and a team of graduate-student collaborators interviewed 200+ businesspeople responsible for setting prices in their firms. In both cases, the central rationale for using interviews was the overabundance of theoretically plausible models of wage or price stickiness, respectively, but with limited ability to distinguish between them via conventional econometric analyses of standard data on labor and product markets. Bewley’s key finding was that concerns about fairness and employee morale figured centrally in explaining why employers do not lower wages when labor-market conditions are slack; at a time when rational forward-looking optimization was the only widely accepted understanding of economic decision-making, Bewley’s work contributed importantly to increasing the credibility of ‘behavioral’ approaches (Howitt, 2002). Blinder *et al.* aimed to gauge empirical support for 12 theories of price stickiness by asking people responsible for setting prices in their firms how well various practices squared with how they made decisions. This resulted in a somewhat complex set of results, in which some theories resonated more with decision-makers than others, but with no one explanation emerging as frontrunner.<sup>11</sup> Both studies generated significant discussion within the discipline about the value of asking economic actors directly about what they do and why, and while they did not pave the way for a wave of similar work, at minimum they established the potential value of well-done qualitative investigation of unresolved topics of major interest.

**Table 1.** Summary of Notable Studies in Economics Using Qualitative and Mixed-Methods Approaches.

| Study                        | Topic                                                                                                                                                       | Qualitative Method and Sample                                                                                                                                                                                                                                                                                            | Value-Added of Qualitative Approach                                                                                                                                                                      |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Macro                        |                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                          |
| Blinder <i>et al.</i> (1998) | Examines 12 competing theoretical explanations for price stickiness, for which empirical validity is hard to establish via traditional empirical approaches | In-depth interviews with a national, multi-industry sample of CEOs, company heads and other corporate price setters. Because each of the 12 theories posits a specific process used by firms to set prices, talking to people should help identify which theory(ies) best describe what firms actually do ( $n = 200$ ). | Possibilities of overcoming deadlocks in standard economic research. Opportunities for considering multiple hypotheses and establishing degrees of empirical support for them                            |
| Bewley (1995, 1999)          | Why don't firms lower wages in recessions?                                                                                                                  | In-depth interviews with businesspeople responsible for making hiring decisions ( $n = 300+$ )                                                                                                                                                                                                                           | Possibilities of overcoming deadlocks in standard economic research. Opportunities for identifying factors not recognized in standard economic thinking (i.e. considerations of fairness and morale)     |
| IO/industry-studies          |                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                          |
| Lerner and Tirole (2000)     | Open-source software                                                                                                                                        | Exploratory study of Apache, Linux, Perl & Sendmail, using written materials, interviews, meetings w/key individuals in each project and highly knowledgeable observers of open source work. ( $n = 4$ )                                                                                                                 | Open-source was new and had not been previously studied. Exploratory study was valuable for characterizing how it works and identifying key economic issues warranting further research.                 |
| Lerner and Merges (1998)     | Determinants of allocation of control rights in biotechnology alliances                                                                                     | Case studies using public securities filings, corporate documents, interviews with senior managers and other observers – combined with regression analysis of quantitative data on biotech alliances.                                                                                                                    | Case studies provided direct evidence supporting a theoretical framework of Aghion & Tirole. Industry contacts confirmed that the regression results provided good characterizations of how things work. |

(Continued)



Table 1. *Continued*

| Study                                 | Topic                                                              | Qualitative Method and Sample                                                                                                                                                          | Value-Added of Qualitative Approach                                                                                                                                                                                                                                                                                 |
|---------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ichniowski, Prennushi and Shaw (1997) | Interrelationships between technology and human-resource practices | In-depth interviews, site visits and analysis of quantitative data for steel plants using the same technology ( $n = 26$ ).                                                            | Demonstrated the importance of complementarities between HR practices and technology in achieving productivity gains.                                                                                                                                                                                               |
| Cockburn & Henderson (1996, 1998)     | Innovation in pharmaceutical industry                              | 'Case histories' of several important drugs, plus in-depth interviews with academic researchers, senior industry researchers and research managers at a number of pharmaceutical firms | Case histories brought out the importance of 'connectedness' to the community of open science as a key factor in driving innovation. Subsequent regression analysis confirmed what was found through the qualitative research.                                                                                      |
| Schwartz (1987)                       | How accurate is the information on which firms make decisions?     | Interviews with small and medium sized metalworking firms in USA, Mexico and Argentina ( $n = 113$ )                                                                                   | Interviews showed that information on policy and technology on which entrepreneurs-based decisions was often poorly aligned with the facts, highlighting the need to drop conceptualizations of businesses as readily acquiring and shrewdly processing information with clear implications for their bottom lines. |
| Autor, Levy and Murnane (2002)        | Adoption of digital check clearing on two floors of a large bank   | Case study approach at a single, large company.                                                                                                                                        | Identified key complementarities between patterns of technological change and changes in skills required to do certain jobs                                                                                                                                                                                         |

*(Continued)*

Table 1. *Continued*

| Study                                   | Topic                                                                                 | Qualitative Method and Sample                                                                                                                                                    | Value-Added of Qualitative Approach                                                                                                                                                                                                                                          |
|-----------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental/contingent-valuation (CV) |                                                                                       |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                              |
| Clark, Burgess and Harrison (2000)      | How do people answer contingent-valuation questions?                                  | In-depth interviews and focus groups ( $n = 15$ ), with subjects recruited from respondents to a CV survey                                                                       | People reported all sorts of trouble answering 'money questions'. Part of the study suggests that CV may not be a solid approach for making decisions about environmental conservation. Some respondents thought participatory public deliberations would be more effective. |
| Desaigues (2001)                        | How readily can people think about and estimate the costs they bear due to pollution? | In-depth interviews with people affected by air pollution ( $n = 73$ )                                                                                                           | It is easier for people to think of costs of pollution in lump-sum terms, rather than as prices, due to the greater simplicity of the cognitive exercise                                                                                                                     |
| Chilton and Hutchinson (2003)           | What are people thinking when they answer WTP questions?                              | In-depth interviews with respondents to a survey about forests in Ireland ( $n = 58$ )                                                                                           | People's rationales for their answers are partly consistent with conventional economic explanations (e.g. diminishing returns) but reflect other motivations as well                                                                                                         |
| Svedsater (2003)                        | What are people thinking when they answer WTP questions?                              | Participants were asked to 'think aloud' about how they were answering questions intended to elicit WTP. Two focus groups of 4 people, plus 21 in-depth interviews ( $n = 29$ ). | Many people do not interpret the valuation questions as intended, although most will not indicate when they don't understand and provide monetary estimates anyway.                                                                                                          |

*(Continued)*

Table 1. *Continued*

| Study                                     | Topic                                                                                                  | Qualitative Method and Sample                                                                                                                                                                         | Value-Added of Qualitative Approach                                                                                                                               |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mulvaney-Day (2005)                       | What value would families having members with serious mental illness place on mental-health treatment? | A 1st phase of cognitive interviews and focus groups ( $n = 23$ ) was used to design WTP questions that were then used in a mail survey of 2000 households (with 840 responding).                     | The qualitative work underlined the emotional impact of the valuation questions for family members and the need to avoid complex probabilities in the scenarios.  |
| Feminist economics                        |                                                                                                        |                                                                                                                                                                                                       |                                                                                                                                                                   |
| Strober, Gerlach-Downie and Yeager (1995) | Child care centers as workplaces                                                                       | In-depth interviews with aides, teachers and directors in large child-care centers in California ( $n = 20$ )                                                                                         | Beneficial to understand childcare centers as workplaces <i>from the point of view of economic agents</i> .                                                       |
| Kim (1997)                                | Evaluation of the Job Training Partnership Act (JTPA)                                                  | Telephone interviews w/open-ended questions. Interviewers were themselves poor women; several had themselves participated in JTPA & so were familiar with it ( $n$ , not reported).                   | Provided multidimensional information into how/why the program did or did affect participants' lives, rather than just measuring effects on wages and employment. |
| Van Staveren (1997)                       | Issues of independence & empowerment in the economic lives of women in Subsaharan Africa               | Focus group among women economists from pan-African network                                                                                                                                           | Illuminated aspects of the research issues that the researcher would not have expected to be important                                                            |
| Olmsted (1997)                            | Education, migration & employment patterns among Palestinian women                                     | In-depth interviews examining how women navigate family pressures and gender norms while still working towards their own goals in education and employment ( $n = 3$ ) (follow-on to standard survey) | Women's strategies for navigating constraints and possibilities were complex, dynamic and multifaceted.                                                           |

*(Continued)*

**Table 1.** *Continued*

| Study                                                        | Topic                                                                                 | Qualitative Method and Sample                                                                                                                                                               | Value-Added of Qualitative Approach                                                                                                                                                                                                            |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Olson and Emami (2002)                                       | Life histories of women economists in the USA                                         | Extended interviews with prominent women economists who received their PhDs in 1950–75 ( $n = 11$ )                                                                                         | Helped develop rich, multifaceted understandings of how the economics discipline changes and improves.                                                                                                                                         |
| King (2011)                                                  | Differences in labor force participation among Mexican women in the USA vs. in Mexico | In-depth interviews ( $n = 18$ )                                                                                                                                                            | Mexican women's much higher labor force participation in the USA reflects different interactions with husbands and different social norms, as well as higher returns to working.                                                               |
| Household saving, spending and borrowing<br>Jefferson (2007) | How women think about saving for retirement                                           | In-depth interviews with women in Western Australia, aiming to capture how they understand what they're doing and why saving-wise ( $n = 30$ ).                                             | Often women viewed saving as a residual – their income flows in, their expenditures are set by their lifestyle and established behaviour patterns within the household and saving occurs if money is left over. Their sense of control varied. |
| Dema-Moreno (2009)                                           | Financial decision-making in Spanish dual-income couples                              | In-depth interviews with a sample of couples differing by age, length of time together and whether or not they had children. Couples were interviewed together and separately ( $n = 16$ ). | Despite claims of making decisions jointly, decision-making patterns often follow established social norms and/or people keep some parts of their finances out of the joint decision-making process.                                           |

*(Continued)*

Table 1. *Continued*

| Study                                         | Topic                                                                                                  | Qualitative Method and Sample                                                                                                                                              | Value-Added of Qualitative Approach                                                                                                                                                                                                                                                               |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kennickell, Starr and Sunden (1996)           | Saving and portfolio behaviours of relatively wealthy households                                       | Focus group with well-off people (annual income > \$250,000 or net worth > \$600,000) ( $n = 8$ )                                                                          | People's 'natural' thinking about how they manage their financial wealth was quite out of line with the standard portrayal of forward-looking deliberation about risks and returns – with much more idiosyncrasy and much less interest.                                                          |
| Social programs, health care and urban issues |                                                                                                        |                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                   |
| Haley, Avery and McMillan (2011)              | How to frame breast-health messages to motivate Appalachian women to attend to their own breast health | Group interviews with Appalachian women ( $n = 77$ )                                                                                                                       | The most effective messages stressed women's roles as care givers and 'the self-perceived reality that the women in this population cannot depend on anyone but themselves'                                                                                                                       |
| Stum (2001)                                   | Factors families consider in thinking about financing long-term care for an elderly member             | In-depth interviews with families coping with paying for a older relative's long-term care ( $n = 45$ )                                                                    | Key factors families consider are: (a) striking the right balance between using and maintaining private financial resources; (b) transferring resources inter-generationally to establish Medicaid eligibility; and (c) deciding not to decide (and accepting the consequences inaction implies). |
| Turney <i>et al.</i> (2006)                   | Effects of experimental program that moved lower income people into better-off neighborhoods           | Mixed-methods study analyzing quantitative data on 636 participants in a randomized control trial, supplemented by in-depth interviews of stratified random sample of 124. | Isolation from networks of information about job opportunities for low-skill workers offset advantages of living in better-off areas.                                                                                                                                                             |

*(Continued)*

Table 1. *Continued*

| Study                                          | Topic                                                                                                                                                                                                                         | Qualitative Method and Sample                                                                                                                                           | Value-Added of Qualitative Approach                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Keels (2008)                                   | How did the Gautreaux program in Chicago – which moved low-income black families from public housing to better-off, racially mixed or white neighborhoods – improve outcomes for children in those families 8–15 years later? | Used both administrative and Census data on neighborhood characteristics, and in-depth interviews with mothers of families who participated in the program ( $n = 25$ ) | Study provided rich insights into children's residential outcomes, participants' experiences of their placement communities, and the processes at work in explaining whether initial gains in neighborhood quality were maintained. The program was in many senses beneficial, even though participants mostly did not feel welcome in their new neighborhoods. |
| Frasure and Jones-Correa (2010)                | What accounts for unexpected collaborations between local governments and community groups aimed at facilitating orderly integration of immigrant groups in suburban areas?                                                   | In-depth interviews among state and local elected and appointed officials and community-based leaders in the Washington, D.C., metropolitan area ( $n = 100$ )          | These alliances: (a) give community organizations access to resources; (b) reduce public agencies' costs of overcoming language and cultural barriers between newcomers and existing residents; and (b) allow local bureaucrats to 'outsource' efforts to nonprofits, while taking credit for the programs, helping them maintain their budgets and staff.      |
| Development & poverty<br>Parker & Kozel (2007) | Dimensions of poverty and vulnerability in Uttar Pradesh and Bihar, India                                                                                                                                                     | In-depth semi-structured interviews in 30 villages, plus survey of 2,250 households.                                                                                    | Factors perpetuating disadvantage among the poor include chronic debt obligations, expensive health shocks, and sparse, localized social capital networks                                                                                                                                                                                                       |

*(Continued)*



Table 1. *Continued*

| Study                                 | Topic                                                                                                           | Qualitative Method and Sample                                                                                                                                                                                                                               | Value-Added of Qualitative Approach                                                                                                                                                                                                                                                                           |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bird, Higgens and McKay (2010)        | Relationships between conflict, education, and the intergenerational transmission of poverty in Northern Uganda | National household survey with in-depth interviews and focus groups in five communities.                                                                                                                                                                    | 'Education supports resilience during and following periods of conflict and insecurity – it is a 'portable' asset of great value'.                                                                                                                                                                            |
| Nikièma and Haddad (2008)             | How do norms of gender relations affect women's access to health care?                                          | In-depth interviews ( $n = 24$ ), focus groups ( $n = 12$ ), and discussions with key informants ( $n = 36$ ) conducted in two ethno-linguistic groups in Burkina Fasso – one with relatively hierarchical and the other more egalitarian gender relations. | Women need to secure the financial and personal support of their husbands to be able to seek health care; this in turn depends on how well they enact expected gender roles. They also have to be able to persuade others that they are notably ill. These factors constitute limits on their access to care. |
| Valente (2011)                        | How land reform in South Africa affected its beneficiaries                                                      | Survey data on 2,279 households, with in-depth interviews and focus groups in 5 communities.                                                                                                                                                                | Qualitative and econometric results show minimal average benefits of participation, with key reason being the poor match between consultant-led land use plans with land grantees' skills.                                                                                                                    |
| Labor<br>Slifman <i>et al.</i> (1999) | Why the growth in performance bonuses, often tied to stock                                                      | In-depth interviews with HR officials at large companies and top HR consulting firms                                                                                                                                                                        | Compensation professions viewed the increased use of performance bonuses as driving by the necessity of following what their competitors were doing.                                                                                                                                                          |
| Cregan (2005)                         | Workers' attitudes towards unions                                                                               | Mail survey in Australia with open-ended questions ( $n = 607$ ).                                                                                                                                                                                           | Inductive analysis sheds light on members' and non-members' own explanations for their membership status Helps generate insights into potentially effective organizing strategies.                                                                                                                            |

*(Continued)*

**Table 1.** *Continued*

| Study                               | Topic                                                                       | Qualitative Method and Sample                                                                        | Value-Added of Qualitative Approach                                                                                                                                                                                                                                                                         |
|-------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Buckland, Fikkert, and Eagan (2010) | What are key barriers to improving capabilities among low-income Canadians? | Financial life histories of low-income people living in Toronto, Vancouver and Winnipeg ( $n = 15$ ) | Longer term stories of people's lives show spells of ups and downs, with 'down' periods dominating. The multiple personal and institutional disadvantages that people faced (mental health, substance problems, incomplete educations) constituted a powerful barrier against improvements in capabilities. |

#### 4.2 *Innovation and Industrial Organization*

Since the 1990s, several high-profile research projects on innovation and IO topics have combined interviews and site visits with collection and analysis of quantitative data. During the 1990s productivity boom, the National Bureau of Economic Research and Sloan Foundation ran a research program on productivity change in industrial companies (see NBER/Sloan, 2000). Some projects in the program focused on dynamic, processual phenomena which would have been virtually impossible to study from standard firm-level quantitative data; for example, Helper (1995) studied how the shift to lean manufacturing in the auto industry affected the organization, efficiency and distribution of risk in the supply chain for auto parts. In a series of studies on the steel industry, Ichniowski and Shaw (with Prennushi, 1997) visited 26 steel plants, conducting interviews and collecting quantitative data on human-resource (HR) practices and productivity. They found important complementarities between technology and HR practices which would have been hard to 'see' without the interviews and plant visits: plants that adopted a package of new human-resource practices (worker teams, greater communication with management, training for multiple jobs, group- and individual-performance bonuses) saw significant gains in productivity, while those adopting new practices in isolation did not. This work has led them to propose an approach to IO research called 'insider econometrics' which combines in-depth investigation of business processes in particular industries with analysis of quantitative data informed by the 'insider' knowledge the researcher acquires. Other IO studies that used case-study methods include Cockburn and Henderson (1996, 1998), whose research on innovation in the pharmaceutical industry highlighted the importance of 'connectedness' to the community of open science as a key factor in driving innovation, and Lerner and Tirole's (2000) exploratory study of open-source software companies.

#### 4.3 *Willingness to Pay and Environmental Economics*

The purpose of contingent valuation (CV) research is to measure how people would value potential benefits of a proposed environmental, health or transportation project, conditional on resource

constraints. This has traditionally been done via closed-end survey questions aiming to elicit people's willingness to pay (WTP) for the potential benefits. Yet as has been known for some time, responses to such questions are hard to trust: they are sensitive to changes in question wording and order, they show odd properties (as when respondents assign the same monetary value to options yielding markedly different material benefits), and they often seem to overstate WTP, among other problems. Thus, a considerable amount of research using qualitative approaches has been done to explore issues underlying these 'anomalies'. A general finding is that, no matter what the wording of a CV question asks respondents to consider, they tend not to narrow down what they think about to only those things. Baker, Robinson and Smith (2008)'s review of qualitative work on this subject identifies four common problems. First, people's 'mental accounting' approaches for thinking about money blur issues that economists expect them to see as trade-offs. Second, they do not necessarily trust that tax revenues earmarked for specific projects will actually be spent as stated. Third, some people object to being asked to put dollar values on things that are 'beyond prices'. And finally, not surprisingly, people tend to get 'warm glow' from expressing concern for the environment and other worthy causes. Qualitative work has also been helpful for identifying ways to get around these problems, as when a first phase of qualitative research is used to figure out how to ask questions in a second quantitative phase. Thus, for example, in Mulvaney-Day's (2005) study of how family members of people with serious mental illnesses would value different treatment options, she first conducted in-depth interviews to gain an understanding about how best to talk with people about these issues. This clarified that emotional impact was a significant issue and that questions using complex probabilities should be avoided. In turn, these findings were used to develop closed-end questions for a mail survey sent to 2000 households.

#### 4.4 *Social Programs, Health Care and Urban Issues*

Mixed-methods research has been used fairly extensively to study effects of social programs on social and economic outcomes for lower income people and their families. This has involved using, in various combinations, administrative records on program participation, traditional closed-end surveys run among large representative samples, in-depth semi-structured interviews conducted in small purposively selected samples, and experiments that randomly assigned program participants to programs with different features. In London, Schwartz and Scott's (2006) valuable review of mixed-methods work evaluating effects of 1990s welfare reforms, they note that qualitative components of this research have been especially valuable for looking into the 'black box' of causal connections between program features and differences in outcomes, because even when experimental design establishes that given types of programs are (or are not) having causal effects, the mechanisms explaining these effects may not be clear. Thus, for example, Turney *et al.* (2006) use qualitative interviews to explore why an experimental program in Baltimore that moved lower income people to better-off neighborhoods failed to improve their employment and earnings outcomes, relative to comparable people who remained in lower income neighborhoods. From the interviews it became clear that both those who moved and those who remained in lower income areas primarily used word-of-mouth to learn of job openings at places that employ workers with relatively low skills; thus, without help developing other strategies for looking for work, people who moved had less access to information on job openings relative to people who remained in lower income neighborhoods, offsetting potential gains of living closer to jobs in better-off places. Had the qualitative interviews not been conducted, it would simply have been unclear why the program failed to improve employment outcomes for those who had moved.

#### 4.5 *'Q-squared' Studies of Poverty in the Developing World*

As mentioned above, there has been a boom in 'q-squared' studies of poverty in the developing world, mostly combining standard closed-end surveys with in-depth interviews, focus groups and/or

ethnographic work (Bamberger, 2000; Kanbur, 2003; Kanbur and Shaffer, 2007). The general premise of the q-squared approach is that, because absolute poverty is a complex and multifaceted phenomenon, characterizing it accurately and designing effective policies and programs to lift people out of it requires research methods that can capture its multifaceted character and dynamics. Much of this research has been conducted with the support of the World Bank and other international development agencies and brings together researchers from different disciplines; some part of it specifically aims to include poor people in the development of characterizations of their lives and problems. One advantage of this work is its ability to unpack how and why given factors affect outcomes, when they may be involved in multiple ways. Thus, for example, Bird, Higgins and McKay (2010) used a variety of qualitative methods in combination with a standard household survey to examine why better-educated people had less adverse outcomes in conflict-ridden Northern Uganda; they found that education mattered most because it improved resilience, enabling people to mobilize resources and find ways out of difficult situations more readily than less-educated people in similar situations. As with program evaluations discussed above, q-squared work is also valuable for explaining why given policies or programs are effective or not. Thus, for example, in studying effects of land reform in South Africa, Valente (2011) combined econometric analysis of survey data collected from a representative survey of 2,279 households with in-depth interviews and focus groups. The quantitative work established that average benefits of participation were quite small, while the qualitative work identified poor matches between land use plans developed by consultants and the skills of land grantees as a key causal factor.

#### 4.6 *Feminist Economics*

Feminist economics has a special concern with expanding the range of empirical methods used in economic research to include methods that can shed light on processes whereby women's viewpoints and economic contributions come to be excluded, disfavored and/or devalued, where a central idea here is that the objectivity of science can improve when new voices are heard and when social problems are examined from the perspectives of the traditionally excluded or oppressed (Harding, 1986). Qualitative work is clearly appealing in this regard, as the open-ended stance facilitates a focus on research subjects' thoughts, experiences, beliefs, etc.; in this sense, it also permits research subjects to participate in the process of building representations of their own economic lives. A study by Kim (1997) illustrates how this kind of approach can have several advantages as well as providing a way for research subjects to tell their own stories. To examine effects of a job-training program for women coming off welfare, Kim trained women who had themselves previously participated in the program to conduct telephone interviews of current program participants. Because these women knew much more about the program and the sorts of lives its participants led than an average interviewer, they were better able to sustain good conversations with interviewees and collect full and accurate information on their experiences. As another example, King (2011) conducted an interview-based study into the question of why Mexican women of given characteristics (age, education, etc.) are much more likely to work for pay if they live in the USA than if they live in Mexico. The rich interview data highlighted that it is not just better returns to working in the USA that induces higher participation, but also quite different interactions with their husbands and the prevalence of different social norms towards women working in the USA.

### 5. Economists' Concerns about the Value of Qualitative Research

Economists tend to express three primary concerns about the value of qualitative research as a means of developing sound empirical characterizations of economic phenomena: that it leaves too much scope for perspectives of the researcher(s) to affect research results; that the quality of self-reported information should be doubted; and that its very richness undercuts its usefulness for building and testing abstract conceptualizations of phenomena of interest. We discuss these three objections in turn.

### 5.1 *Perspectives of the Researcher(s) may Affect Research Results*

In general, economists take seriously the physical-science ideal of ‘good science’, whereby faith is put only in findings that are ‘objective’ and in no way depend on the specific measurement apparatus used (see e.g. McCloskey, 1983: 482–483; Helper, 2000; Dow, 2002, chap. 5). Taken literally, this seems to imply that open-ended research cannot be good science, as its conduct seems to depend integrally on judgmental decisions of individual researchers, whereas closed-end research seems to narrow the scope for variations across researchers in what information is collected and how it is interpreted. This sort of contrast, while oversimplified, does correctly underline that, to be valuable, knowledge generated from a given research program should be based on rigorous and systematic efforts to represent the phenomena of interest in a faithful way, that is, with sufficient accuracy, sufficient incorporation of major dimensions of variation or complexity, and sufficient understanding of dynamic elements, so that decisions, programs or interventions based on the representation would permit better outcomes than would have been possible had the research not been done. But this objection overlooks the fact that well-done qualitative work typically uses a variety of practices to reduce odds of results depending in undue ways on specifics of research methods and/or characteristics of researchers. These include: (a) fully explaining and documenting procedures and protocols, so that the methodology of the study is transparent and potentially replicable; (b) explaining in the write-up any unique aspects of the researcher or the research design that might make the results different from what another study using a similar method would find; (c) conducting research in teams, often interdisciplinary, to benefit from exchanges among researchers with different perspectives; and (d) making systematic use of opportunities to cross-check findings via other information sources (i.e. triangulate), so that those which do cross-check can be understood as having better empirical support than those which do not.<sup>12</sup> Thus, while individual studies may do more or less well in establishing that their results faithfully represent the phenomenon of interest, well-done qualitative work aims specifically to attenuate potential problems or issues of bias due to perspectives of individual researchers.

### 5.2 *The Quality of Self-Reported Information Is Potentially Problematic*

Economists tend to be skeptical that asking people directly about what they do and why yields good information, for two sets of reasons.<sup>13</sup> The first and more straightforward of the two concerns the fact that people may have incentives to misrepresent information about themselves to researchers. They may underreport behaviours they are embarrassed about, yet over-report socially desirable ones. They may tend to ‘rewrite’ aspects of their personal histories in ways that reduce their responsibility for bad events and attach the blame to others. They may overstate how much they would benefit from gaining access to a good or service or how much they are harmed by not getting access to it, and so forth. However, these sorts of problems, while tricky, can often be tackled through the research project’s design and methodology, and do not constitute *a priori* reasons to disbelieve that self-reported information could ever provide meaningful or valuable insights into the phenomenon of interest. There is a long tradition of studying ‘response problems’ in survey research, which provides a considerable amount of guidance on how to ask questions in ways that elicit high-quality responses, and identifies circumstances under which response quality is most likely to be problematic (e.g. Schaeffer and Presser, 2003; Bradburn, Sudman and Wansink, 2004). For example, if sensitive questions (e.g. about finances, intimate matters, or illicit behaviours) are asked early in the interview, without first building a rapport of trust between the interviewer and the respondent, then it is indeed likely that the respondent will refuse to answer or answer in a way that misrepresents his situation. But again, many well-known practices can be used to minimize misreporting and non-reporting on sensitive issues, including: providing considerable assurances of confidentiality and anonymity; explaining the high value of the respondent’s truthful answers for the success of the research; asking non-threatening questions before

turning to sensitive issues; training the interviewer to maintain a nonjudgmental stance; and/or asking the respondent to enter her answers directly into a computer. Thus, just as in closed-end research, careful research design can go a long way towards attenuating response problems in open-ended work.<sup>14</sup>

The second set of concerns about self-reported information relates to possibilities that people may or may not know or be readily able to articulate what the researcher wants to know about. This relates to Milton Friedman's analogy of the billiards player, who has no formal knowledge of principles of physics but plays 'as if' he does; in this case, a model of the player's behaviour built from his own descriptions of how he plays could be much less valuable for making predictions than one that assumes he plans his shots using principles of physics. In fact, it *is* important to establish that one's research subjects have the knowledge, information, perspectives, experiences and interest in the topic that will enable them to serve as good 'key informants' with respect to the issues of interest. As an example when this was not the case, Slifman *et al.* (1999) tried to investigate why compensation for executives and top professionals was shifting away from salaries and towards bonuses and stock options in the 1990s, by interviewing HR executives at large companies and at consulting firms specialized in setting corporate compensation. Respondents tended to report that their companies were adjusting their compensation practices because their competitors were; very few had much insight into what was driving the change at the market level. While this underlines the importance of conducting initial rounds of interviews, to make sure that people expected to be able to serve as key informants can actually do so effectively, it does not imply that talking to people is inherently problematic as a means of trying to advance economic knowledge. Rather, the issue is to make sure that the research question is well matched with research subjects' ability to shed insight into it, and/or that other complementary sources of information can be used to round out the kinds of information and insights they can offer, as in case studies.<sup>15</sup>

### 5.3 *Qualitative Information Is Fine for Description, but Not for Explanation*

A final concern about qualitative information is that it fits poorly into the standard hypothetico-deductive method taken to be the preferred way of making major 'truth claims' in economics in the past 50 or so years (McCloskey, 1983, 1998; Backhouse, 2007). Via this method, a conceptual framework is developed from first principles, testable implications are derived from it, then quantitative data are used to test these implications statistically. Given the small, unrepresentative samples typical of qualitative research and the multifaceted character of the information collected, it is usually ill suited to this sort of project. This gives rise to the claim that qualitative research can be fine for description, but not for explanation.

But again, this is not an issue of inherent characteristics of qualitative work, but rather of the design of specific qualitative projects. As is clear from discussion above, some qualitative work is exploratory and descriptive, aiming to pave the way for further work in the area (e.g. Lerner and Tirole's 2000 study of open-source software). But other studies are clearly 'explanatory' in orientation, for example, aiming to gauge empirical support for existing theories (e.g. Blinder *et al.*, 1998), or explain why field experiments fail to have expected causal effects (Turney *et al.*, 2006), or identify causal relationships not readily deducible from first principles (Cockburn and Henderson, 1996, 1998), etc. What is different in qualitative research projects is that, rather than setting up clear tests of hypotheses that result in zero/one judgments about empirical support, efforts to gauge the validity of theories or characterize causal relations tend to follow more of an 'informal Bayesian logic' (Bennett, 2004), whereby researchers begin with a set of working hypotheses about the phenomenon of interest, then revise their ideas as they encounter new information.<sup>16</sup> The end result then is typically not zero/one judgments about initial hypotheses, but rather sets of characterizations and explanations that have been revised and modified according to what was found in the field. While there is a good



amount of discussion presently about practices that can be used to improve the rigor with which this is done,<sup>17</sup> the expanding body of work in this area illustrates that, rather than being suitable for descriptive purposes only, qualitative and mixed-methods work often has good potential for identifying and characterizing causal processes, even if we are still in the process of figuring out best ways to realize it.

## 6. Concluding Remarks about Research Ethics and the Voices of Research Subjects

This paper has argued that well-done qualitative work can provide scientifically valuable and intellectually helpful ways of adding to the stock of economic knowledge, especially when applied to research questions for which they are well suited. With the economics discipline increasingly willing to examine key assumptions about behaviours of individuals and interactions between them (e.g. Camerer, 1999; Henrich *et al.*, 2004; Fuster, Laibson and Mendel, 2010; Kahneman, 2011), it is a particularly propitious time to be thinking of qualitative methods as a potentially valuable instrument in the economists' toolkit, where collaboration with researchers from other disciplines can be valuable for overcoming economists' lack of training in their use.

A final point concerns issues of research ethics that arise in thinking about benefits of increasing the use of qualitative methods in economic research. As has been discussed by Ruccio (2008) and DeMartino (2011), theoretical and empirical representations developed by economists *matter* for those whose behaviours they are intended to capture, in so far as such representations inform decisions that have material consequences for people's incomes, employment, access to public services or social insurance, retirement security, etc. And yet, economists' research subjects (households, firms, traders, governments, etc.) typically have little voice in the construction of representations that affect them: rather, it is assumed *a priori* that advanced training in economic theory and methodology and strong command of existing scholarly literature are required to be able to contribute to economic knowledge, so that the thoughts, beliefs and insights of laypeople will usually have limited value. Thus, some have questioned whether research methods that systematically exclude the voices of the researched are consistent with sound principles of research ethics, including respecting the dignity and autonomy of research subjects (McGee, 2003); this is the line of thinking that gave rise to the World Bank's influential study, *Voices of the Poor* (Narayan *et al.*, 2000), and underpins the stated commitment of the World Bank and IMF to participatory methods of assessing social groups' needs and devising good programs, policies and projects to meet them (World Bank, 2000). Others have pointed out that disregarding perspectives of research subjects is potentially inconsistent with a 'stakeholder' approach to building knowledge, in which all people whose lives may be affected by decisions following from this knowledge should have an opportunity to participate in its construction (see, for example, Dench, Iphofen and Huws, 2004).

In this respect, well-designed qualitative projects, by virtue of their open-ended approach to gathering information, provide valuable avenues for bringing perspectives of actual economic actors more directly into the processes of producing economic knowledge. Research projects like that of Bewley demonstrate that bringing the perceptions, experiences and understandings of research subjects into discussions of economic issues and topics that concern them can yield unexpected and highly valuable insights. The development of means of archiving qualitative and mixed-methods data sets, so as to permit broader access to research material while also protecting the confidentiality of research participants, can be especially valuable for validating results of individual studies and opening up the process of exploring and interpreting research findings.<sup>18</sup> Potentially, then, making greater use of research strategies that give economic agents more opportunities to help shape how economic knowledge evolves may result in knowledge that has better scientific validity, higher social value and better ethical properties than what the discipline has produced to date.

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## Notes

1. In other social sciences and in the research-methodology literature, an enormous amount has been written about this distinction; see, for example, Hulme (2007) and references therein. For additional discussion of distinctions between qualitative and quantitative research relevant to economic research, see Morgan and Smircich (1980), Nelson (1995), Helper (2000), Kanbur (2003), Carus and Ogilvie (2009), and Chamlee-Wright (2010).
2. Kanbur (2003) identifies five key dimensions of difference between qualitative and quantitative research, although they can be seen as corollaries of the open- vs. closed-end distinction rather than essential features of each research type: (1) non-numerical vs. numerical information, (2) specific to general population coverage, (3) active to passive population involvement, (4) inductive vs. deductive methodology, and (5) broad social science vs. neoclassical economics as the disciplinary framework. See also Kanbur and Shaffer (2007) for discussion of 'incidental' vs. 'essential' differences between qualitative and quantitative approaches.
3. For discussions of triangulation, see Jick (1979), Flick (1991), Olsen (2003), Downward and Mearman (2007), and Carus and Ogilvie (2009).
4. See Bewley (2002) for discussion of in-depth interviews as a tool of economic research.
5. See, e.g. Dewey (1910), McGoun (1936), and Bergmann (2005, 2007).
6. Also in the category of case-based methods is Qualitative Comparative Analysis (QCA), which uses logic-based analysis to identify causal relations from relatively small numbers of observations (Ragin 1987). QCA research with important economic dimensions includes Ragin (1994) on pensions, Dy (2005) on health-services research, Krause (2009) on finance ministries and the budget process, Castellano (2010) on broadband adoption, Lam and Ostrom (2010) on development interventions and Crawford (2012) on energy planning.
7. Bergman (2008) and Creswell and Plano Clark (2010) are valuable references on the methodologies of mixed-methods research.
8. See Paluck (2010) for discussion of potential gains from combining qualitative and experimental research.
9. See, for example the collection of papers in Denzin and Lincoln (2011).
10. Chamlee-Wright (2010) is an example of economic research using software-facilitated compilations of word mentions as primary means of analyzing qualitative information.
11. Theories receiving the most support were those emphasizing 'coordination' issues (not wanting to raise prices because competitors might not follow), varying non-price characteristics of products to handle small changes in demand or costs, and informational and lag issues in passing through changes in costs. Theories receiving the least support were those emphasizing fears that customers would confuse price cuts for reductions in quality, properties of the marginal cost curve, and/or hierarchical decision-making within firms (see Blinder *et al.*, chap. 5).
12. A disadvantage of using these practices intensively is that they can result in papers that are longer and more detailed than standard journal articles, eroding their publication prospects; a suggestion here is to post detailed explanations of procedures and protocols on the internet to which readers can be referred (National Science Foundation, 2004).



13. See Beshears *et al.* (2008) for discussion.
14. In fact, because interviewers can probe and cross-check people's answers in open-ended interviews, they provide extra scope for reducing response problems. Thus, for example, qualitative interviewers are often trained to spot and correct potential problems with misreporting; for example, if a respondent says she did x because of y, her view of the causality can be checked by asking her to speculate on what would have happened x-wise had y not happened (NSF, 2004).
15. As discussed in Downward and Mearman (2007), it is to be expected that given social phenomena may look different from different perspectives; this is precisely the value of using multiple methods to investigate a given phenomenon, then triangulating what is learned from them. See also Tashakkori and Teddlie (2010) for discussion of the philosophical underpinnings of mixed-methods research.
16. Buckley (2004) specifically discusses application of Bayesian method in the conduct of qualitative research. See also Poirier (1988, 2006) and Sims (2007).
17. See Huberman and Miles (1985); King, Keohane and Verba (1994); Maxwell (2004); and NSF (2004) and references therein.
18. See, for example, U.K. Data Archive (2010) or U.S. National Science Foundation (2012). In turn, making qualitative-research materials widely available poses relatively tricky ethical issues related to consent and confidentiality (see, e.g. Corti, Day and Backhouse, 2000).

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